

WEEK3-BOOKFLOW(MONGO DB):

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (4)' panel lists the 'BOOKFLOW' database and its collections: 'book_metadata', 'admin', 'config', 'local', 'studentdb', 'localhost:27017', and 'StudentCollection'. The main editor area shows a JavaScript console with the following code and output:

```
> use("BookFlow")
< switched to db BookFlow
> db["book_metadata"].find()
<
> use BookFlow
< already on db BookFlow
> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: new Date("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: new Date("2023-02-20")
    }
  ]
});
< {
```

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (4)' panel lists the 'BOOKFLOW' database and its collections: 'book_metadata', 'admin', 'config', 'local', 'studentdb', 'localhost:27017', and 'StudentCollection'. The main editor area shows a JavaScript console with the following code and output:

```
> use("BookFlow")
< switched to db BookFlow
> db["book_metadata"].find()
<
> use BookFlow
< already on db BookFlow
> db.book_metadata.insertMany([
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: new Date("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: new Date("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
  }
]);
< {
  acknowledged: true,
  insertedIds: ObjectId('6abf972b969855a01bd76244')
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

- BOOKFLOW
 - BookFlow
 - book_metadata
 - admin
 - config
 - local
 - studentdb
- localhost:27017
- localhost:27017
- StudentCollection

```
>_MONGOSH
{
  title: "Whispers of Yesterday",
  author: "S. Kapoor",
  genre: "Fiction",
  publishedYear: 2019,
  formats: ["Hardcover", "Paperback"],
  specs: {
    pages: 410,
    binding: "Sewn"
  },
  avgRating: 3.8,
  reviews: [
    {
      member_id: "M3801",
      rating: 3,
      comment: "Slow start but strong ending.",
      date: new Date("2020-01-15")
    },
    {
      member_id: "M3802",
      rating: 4,
      comment: "Beautifully written prose.",
      date: new Date("2020-02-11")
    }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: "P. Iyer",
    title: "Quantum Horizons"
  }
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

- BOOKFLOW
 - BookFlow
 - book_metadata
 - admin
 - config
 - local
 - studentdb
- localhost:27017
- localhost:27017
- StudentCollection

```
>_MONGOSH
{
  member_id: "M4801",
  rating: 5,
  comment: "The Digital format made annotations so easy.",
  date: new Date("2023-08-01")
},
{
  member_id: "M4802",
  rating: 5,
  comment: "Mind-bending and well narrated.",
  date: new Date("2023-09-14")
}
]
});
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6abf973f969055a01bd76245'),
    '1': ObjectId('6abf973f969055a01bd76246'),
    '2': ObjectId('6abf973f969055a01bd76247')
  }
}
> db.book_metadata.find(
{
  avgRating: {
    $gt: 4
  }
}
);
< {
  _id: ObjectId('6abf972b969055a01bd76244'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: {
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

BOOKFLOW

- BookFlow
 - book_metadata
- admin
- config
- local
- studentdb

localhost:27017

localhost:27017

StudentCollection

```
>_MONGOSH
{
  'Audobook':
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  }
}
{
  _id: ObjectId('6abf973f969055a01bd76245'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

BOOKFLOW

- BookFlow
 - book_metadata
- admin
- config
- local
- studentdb

localhost:27017

localhost:27017

StudentCollection

```
>_MONGOSH
genre: 'Technology',
publishedYear: 2021,
formats: [
  'Digital',
  'Paperback'
],
specs: {
  pages: 210,
  fileSize_MB: 2.1,
  drm: false
},
avgRating: 4.2,
reviews: [
  {
    member_id: 'M2001',
    rating: 4,
    comment: 'Solid Digital read, clean layout.',
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
}
{
  _id: ObjectId('6abf973f969055a01bd76247'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

- BOOKFLOW
 - BookFlow
 - book_metadata
 - admin
 - config
 - local
 - studentdb
- localhost:27017
- localhost:27017
- StudentCollection

```
>_MONGOSH
}
> db.book_metadata.find(
  {
    formats: {
      $in: ["Digital", "Audiobook"]
    }
  }
);
< [
  {
    _id: ObjectId('6abf972b969055a01bd76244'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ],
  }
]
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

- BOOKFLOW
 - BookFlow
 - book_metadata
 - admin
 - config
 - local
 - studentdb
- localhost:27017
- localhost:27017
- StudentCollection

```
>_MONGOSH
  {
    date: ISODate('2022-05-10T00:00:00.000Z')
  },
  {
    member_id: 'M2002',
    rating: 5,
    comment: 'Best programming book this year.',
    date: ISODate('2022-06-01T00:00:00.000Z')
  }
]
{
  _id: ObjectId('6abf973f969055a01bd76247'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ],
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

BOOKFLOW

BookFlow

book_metadata

admin

config

local

studentdb

localhost:27017

localhost:27017

StudentCollection

Welcome

mongosh: BOOKFLOW

>_MONGOSH

```
      formats: {
        $regex: "Digital",
        $options: "i"
      }
    }
  };
}
< {
  _id: ObjectId('6abf972b969055a01bd76244'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: ISODate('2023-01-15T00:00:00.000Z')
    },
    {
      member_id: 'M1042',
      rating: 4,
```

Compass

My Queries

Data Modeling

CONNECTIONS (4)

Search connections

BOOKFLOW

BookFlow

book_metadata

admin

config

local

studentdb

localhost:27017

localhost:27017

StudentCollection

Welcome

mongosh: BOOKFLOW

+

> MONGOSH

```
]
}
{
  _id: ObjectId('6abf973f969055a01bd76247'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [
    'Digital',
    'Audiobook'
  ],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: 'P. Iyer',
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
}
BookFlow>
```